SEC 3 · LEC 3 · INTERPLANETARY TRANSFERS + FLYBYS

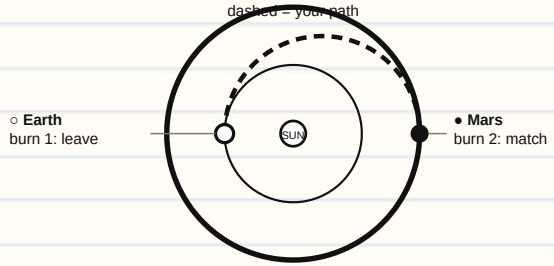
Lazy throws and bouncing off moving planets

Left = your left notebook page. Right = your right page. Draw both panels before you look closely at mine.

LEFT PAGE · DRAW

S3 / L3 — Transfers & flybys **date · pg __**

1 · HOHMANN = the lazy throw

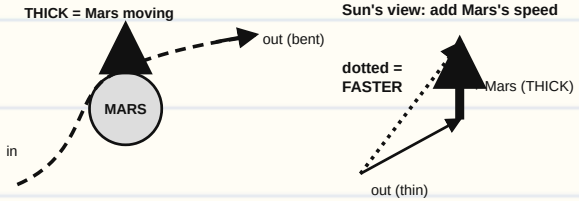


Half an ellipse, 180° around the Sun. Cheapest Δv.
Earth → Mars ≈ 8.5 months = half the ellipse's period (your S1 quiz).

LAMBERT'S PROBLEM = the "can I get there by then?" calculator
give: 2 positions + a travel time → get: the one orbit linking them → read off the Δv
Mars in 20 days? The orbit exists. The Δv is absurd. The rocket can't be built. → NO
fast = expensive · slow = cheap · more speed at the same spot → bigger orbit → reach farther

2 · FLYBY = bounce off a moving planet

Mars's view: speed in = speed out, only the direction bends.



✏️ **PENCIL KEY** · thin solid = Earth's orbit · thick solid = Mars's orbit · dashed = your path · ○ hollow dot = start · ● filled dot = target · THICK arrow = the planet moving · dotted arrow = the result

⭐ **3 TAKEAWAYS** 3 lines

1. Getting there fast costs fuel; the Hohmann half-ellipse is the cheapest, slowest sensible path
2. A flyby bends your path around a planet and, because the planet is moving, hands you some of its speed
3. More speed at the same spot = bigger orbit = farther reach; the price of flybys is years of travel

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
Lambert's problem: what do you give it, what does it give back?	transfer = move from one orbit around the Sun (Earth's) to another (Mars's)
Why can't we fly to Mars in 20 days?	Lambert's problem: give 2 positions + a travel time → math returns the ONE orbit linking them → tells you the Δv
Hohmann: what shape, what angle, how many burns, why cheapest?	fast = expensive: Mars in 20 days needs absurd Δv → rocket too big → not doable
How long is a Hohmann to Mars? What's it got to do with the S1 quiz?	Hohmann = half-ellipse touching both orbits · depart/arrive 180° apart · 2 burns · cheapest Δv (= your perigee/apogee burn line, around the Sun)
Flyby: in Mars's eyes does the speed change? In the Sun's eyes? Where does the extra come from?	Earth→Mars Hohmann ≈ 8.5 months · TOF = half the ellipse's period (S1 quiz) · window every 26 months
More speed at the same spot → what happens to the orbit?	Hohmann isn't required, it's the cheapest · real choice = trade: time vs fuel
Why did Rosetta take 10 years?	far targets (Jupiter+): Hohmann Δv still too big → flyby / gravity assist instead
? can a flyby SLOW you down	flyby: Mars's view → speed in = speed out, direction bends · Sun's view → add Mars's own speed → faster (pass the other side → slower)
	more speed at same spot → bigger a → reach farther · energy stolen from the planet (it slows by nothing you could measure)
	Rosetta (ESA): launch 2004 → Earth '05, Mars '07, Earth '07, Earth '09 → comet 2014 · cost of flybys = TIME

SEC 3 · LEC 3 · TEACH + FACT CHECK

You don't aim at Mars. You aim at where Mars will be.

Start with the box. Then the three ideas, the caption errors, and where the instructor was loose.

🧒 Like you're 5

You're on a merry-go-round and your friend is on a bigger one around the same pole, going slower. You want to throw them a ball. You could hurl it straight at them, hard, but you'd have to throw *really* hard and it'd arrive going way too fast to catch. Or you could lob it gently so it drifts outward on a long curve and arrives just as your friend comes around. The gentle lob takes longer but barely any effort. That lob is a **Hohmann transfer**, and it's how we send things to Mars.

Now the second trick. If a ball bounces off a wall, it comes back at the same speed. But if it bounces off the **front of a moving truck**, it comes back faster, because the truck gave it a push. Planets are moving trucks. Swing a spacecraft close past one and it comes out faster than it went in. That's a **flyby**, and it's how we reach Jupiter and beyond without a rocket the size of a city.

🎯 The three ideas

- **Lambert's problem** is the "can I get there by then?" calculator. You hand it where you are, where you want to be, and how long you're willing to take. It hands back the one orbit that connects those two points in that time, and from the orbit you read off how much speed change (Δv) you'd need. Ask it for Mars in 20 days and it happily gives you an orbit, but the Δv is tens of km/s, which no rocket can carry. The math works; the fuel doesn't exist. Fast is expensive. Slow is cheap.
- **The Hohmann transfer** is the cheapest answer: half an ellipse that just kisses Earth's orbit on the inside and Mars's orbit on the outside. You leave Earth at one end and arrive at Mars at the other, 180° around the Sun. Two burns: one to leave, one to match Mars. This is exactly your line from last lecture, scaled up to the Sun: burn at the low point to raise the high point. For Mars it takes about 8.5 months, which is half the period of that ellipse, which is the time-of-flight formula from your very first quiz. And because Mars has to be in the right place when you arrive, you can only launch every 26 months.
- **The flyby / gravity assist.** Seen from Mars, a passing spacecraft comes in at some speed and leaves at the same speed, just bent in a new direction. Seen from the Sun, though, Mars itself is moving at 24 km/s, and the bend lines the spacecraft up to carry some of that along. More speed at the same distance from the Sun means a bigger orbit (your semi-major axis grows), which means you can reach farther out. The energy comes from Mars, which slows down by an amount nobody could ever measure. Pass behind the planet to speed up; pass in front to slow down (that's how you brake at Mercury).

⚡ Say it as: fast costs fuel, Hohmann is the lazy throw, flybys

🔎 Caption errors

Captions said	Instructor meant
"human / home and / common / whole and transfer"	**Hohmann** transfer (every time)
"their limits problem"	**Lambert's** problem
"a connex approximation"	a **conic** approximation (ellipses, hyperbolas)
"velocity of rotation of mass"	velocity of **Mars**
"destroying"	"this drawing"
"the opposite"	the **orbit**
"flew to a comet in 2010"	launched 2004, arrived at the comet 2014 (2010 was an asteroid flyby on the way)
"get to an asteroid"	get to a **comet**

▲ Where the instructor was loose

- **"Lambert's problem takes into account the Δv and the time."** Backwards. Time (and the two positions) go *in*; the orbit and its Δv come *out*. Small, but it's the difference between understanding the tool and just naming it.
- **"Hohmann is the most efficient transfer."** True for almost every real case. (For a target more than ~12× farther out, a three-burn path can beat it. You'll never need this; just know "almost always.")
- **"Velocity and semi-major axis are related directly."** Right idea, right conclusion. The precise statement is that orbital *energy* sets the size, and more speed at a given spot means more energy. Same thing in practice.

SEC 3 · LEC 3 · VOCABULARY

Every word used, in plain English

Two people's names are in here (Hohmann, Lambert). They're just labels on ideas. Copy only the new words into the word bank.

Transfer / transfer orbit — the path you fly to get from one orbit to another, like from Earth's orbit around the Sun to Mars's. *The highway between two parking lots.*
Interplanetary — between planets, orbiting the Sun rather than Earth. *Repeat.*
Lambert's problem — the classic "find the orbit" puzzle: given a start point, an end point, and a travel time, what orbit connects them? Named after Johann Lambert, an 18th-century mathematician. *The "can I get there by Tuesday?" calculator.*
Δv (delta-v) — the total speed change a trip requires, which is really the fuel bill. *Repeat. Still the currency of spaceflight.*
Propulsion / propellant — the engine, and the fuel it burns. "Not enough propulsion" means "can't carry enough fuel to make that Δv."
Input / constraint — what you feed into a calculation / a limit you can't break (rocket size, fuel mass, time). *Repeat.*
Conic / conic approximation — treating every orbit as one of the shapes you get by slicing a cone: circle, ellipse, parabola, hyperbola. "Patched conics" = stitching an Earth-centered ellipse to a Sun-centered one to a Mars-centered hyperbola. *Lego bricks of orbit shapes.*
Hohmann transfer — the half-ellipse path that touches both the starting and ending orbit, with departure and arrival 180° apart and two burns. Cheapest in fuel. Worked out by Walter Hohmann, a German engineer, in 1925. *The lazy throw.*
Tangent — just touching, not crossing. The Hohmann ellipse is tangent to both circles, which is why the burns are small: you're already moving along the right line.
Optimal / efficient / most efficient — in this lecture, always means "least Δv," i.e. least fuel. Not fastest.
Time of flight (TOF) — how long the trip takes. For a Hohmann it's half the ellipse's period. *Your Section 1 quiz question.*
Launch window / synodic period — the stretch of days when the planets are lined up right for a Hohmann. Earth–Mars: about every 26 months. *Repeat from S1 primer.*
Phase angle — how far ahead the target planet must be when you launch so it's there when you arrive. Mars: ~44° ahead. Not on the slide; it's why the window exists.
Flyby / gravity assist / slingshot — three names for one trick: swing close past a planet so its gravity bends your path and, because the planet is moving, changes your speed relative to the Sun. *Bouncing off the front of a moving truck.*
Gravity field — the region around a planet where its pull matters. It's what bends the flyby path.
Relative velocity — your speed as seen from something else (Mars, say) rather than from the Sun. The flyby doesn't change your speed relative to Mars, only relative to the Sun. *Repeat.*
Vector — an arrow: a quantity with both a size and a direction. Velocity is a vector. The slide's arrow pictures are vector addition: spacecraft-relative-to-Mars plus Mars-relative-to-Sun equals spacecraft-relative-to-Sun. *Two pushes added tip to tail.*
Magnitude / direction — the length of the arrow / which way it points. A flyby changes the direction in Mars's frame and the magnitude in the Sun's frame.
Hyperbola — the open, swing-past shape of a flyby path near a planet ($e > 1$). *Repeat.*
Semi-major axis (a) / "mars energetic orbit" — size of the orbit. Mars speed at the same spot — more energy

SEC 3 · LEC 3 · SAY IT OUT LOUD

Page 4: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I fill the box.

🗣 The friend's opening question

"How do we even get to Mars? Don't you just point the rocket at it and go?"

✅ Three sentences you should be able to say

- No, you aim at where Mars is going to be. The cheapest way is a long lazy curve that drifts outward and meets Mars halfway around the Sun, about eight and a half months later.
- You could go faster, but every bit faster costs a huge amount more fuel, and past a point the rocket just can't be built. Fast is expensive, slow is cheap.
- For really far places like Jupiter, you cheat: you swing close past a planet and it slings you out faster, like bouncing a ball off the front of a moving truck. It costs no fuel, just years of travel.

❓ Follow-ups a friend might ask

- "Why can't you just go whenever?" (Mars has to be in the right spot when you arrive; the lineup comes every 26 months)
- "Where does the extra speed from a flyby come from? Isn't that free energy?" (from the planet; it slows by an unmeasurable amount)
- "Can a flyby slow you down?" (yes, pass in front of the planet instead of behind)
- "So why did it take Rosetta ten years?" (four flybys to build up speed for free instead of carrying fuel; time was the price)
- "What's Lambert's problem?" (a calculator: two points plus a time, gives you the orbit and the fuel)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Section 3 is done after this. Next: the Section 3 conversation test, five questions from me, party-style.